Geocomputation

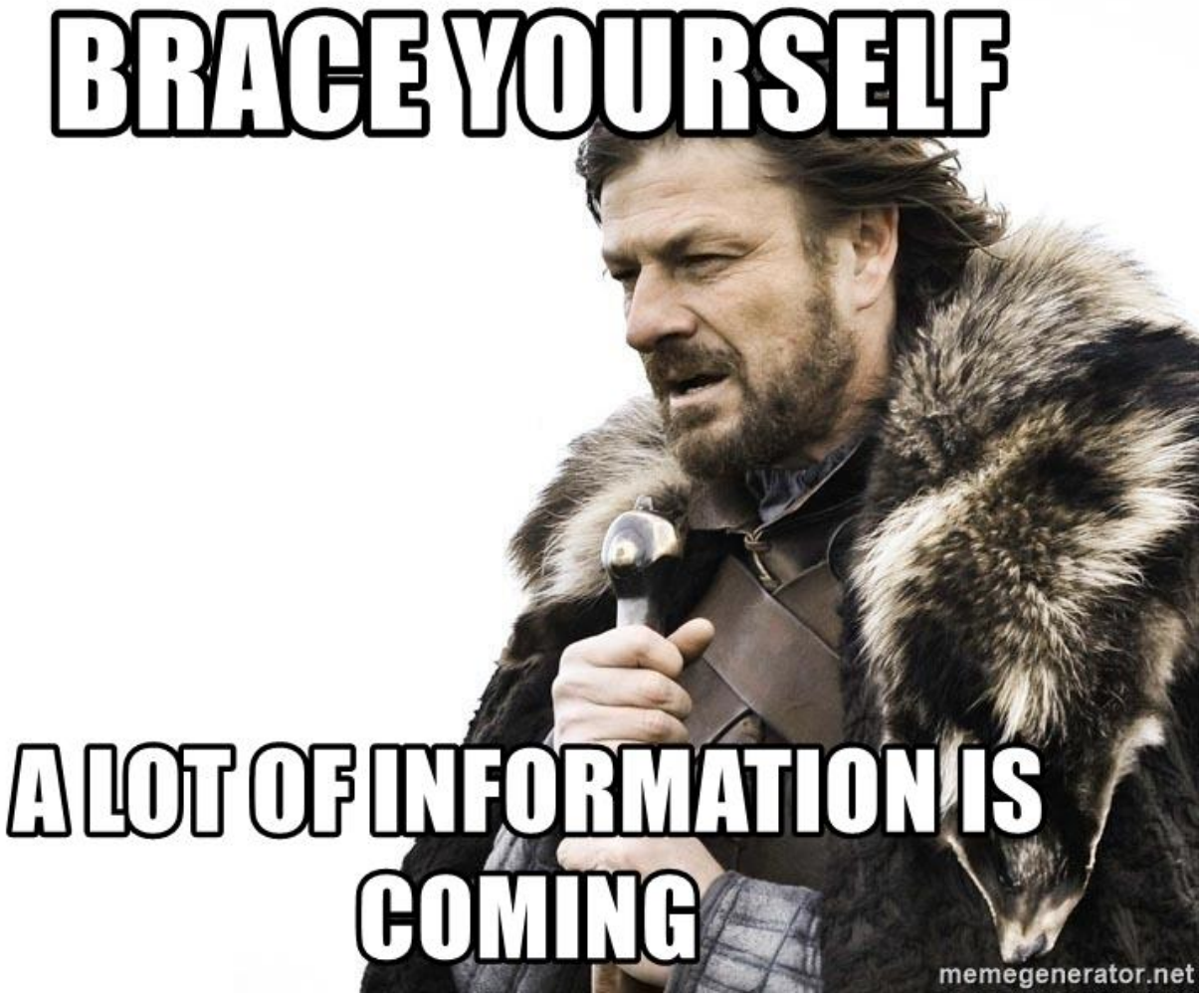
Beyond the Choropleth



Module outline

W1	Reproducible Spatial Analysis	Core Spatial Analysis
W2	Spatial Queries and Geometric Operations	
W3	Point Pattern Analysis	
W4	Spatial Autocorrelation	
W5	Spatial Models	
W6	Raster Data Analysis	Applied Spatial Analysis
W7	Geodemographic Classification	
W8	Accessibility Analysis	
W9	Beyond the Choropleth	Data Visualisation
W10	Complex Visualisations	

This week


BRACE YOURSELF
A LOT OF INFORMATION IS COMING
memegenerator.net

This week

Part I: Positioning the map

- Coordinate reference systems and projections.

Part II: Making maps

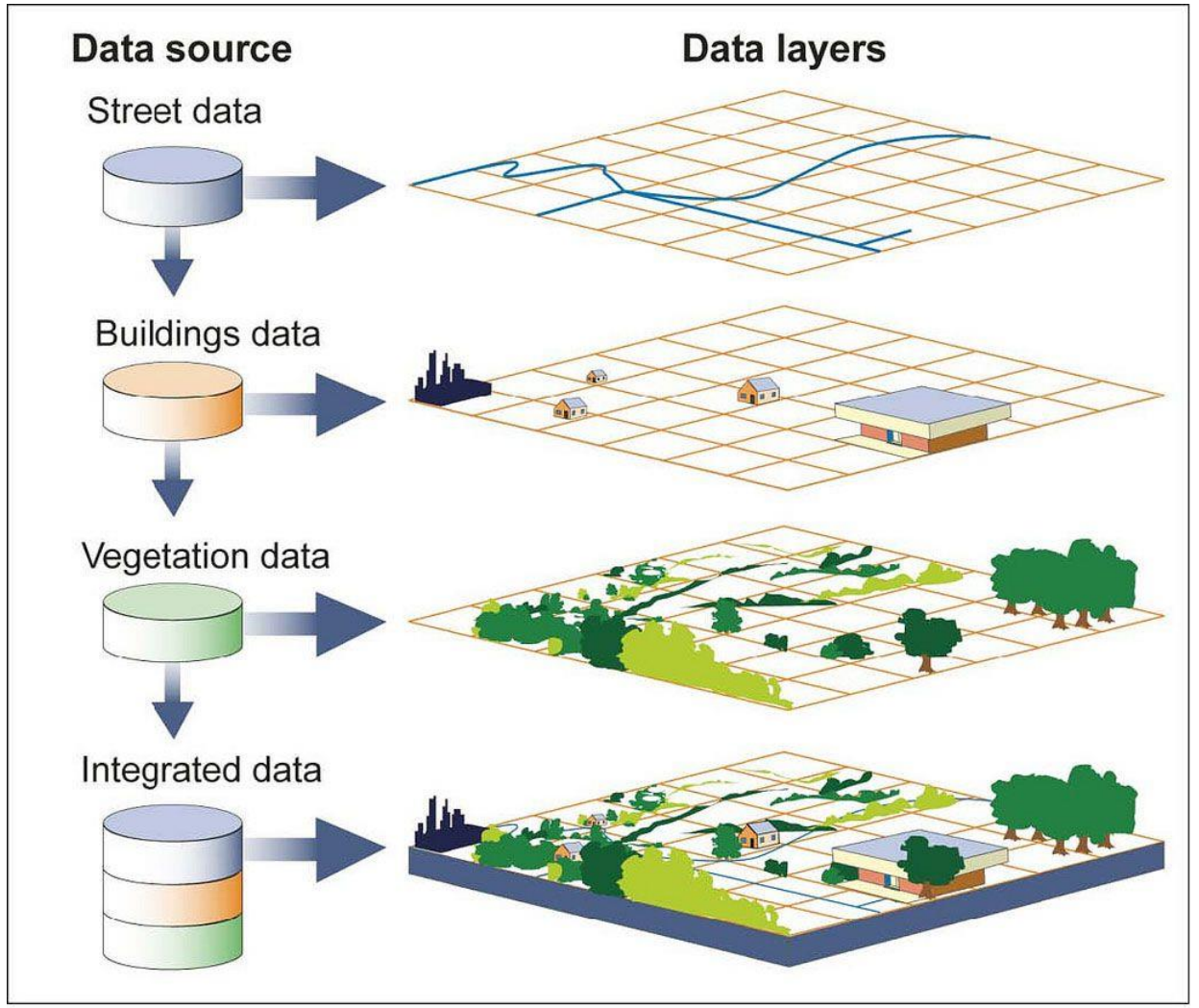
- Choropleth maps: classification, standardisation and colour.
- Bivariate maps and changing boundaries.
- Map conventions.

Part III: Working with an AI assistant

- Checking what a tool produces.

Positioning the map

A GIS stacks layers that must line up



Source: GAO.

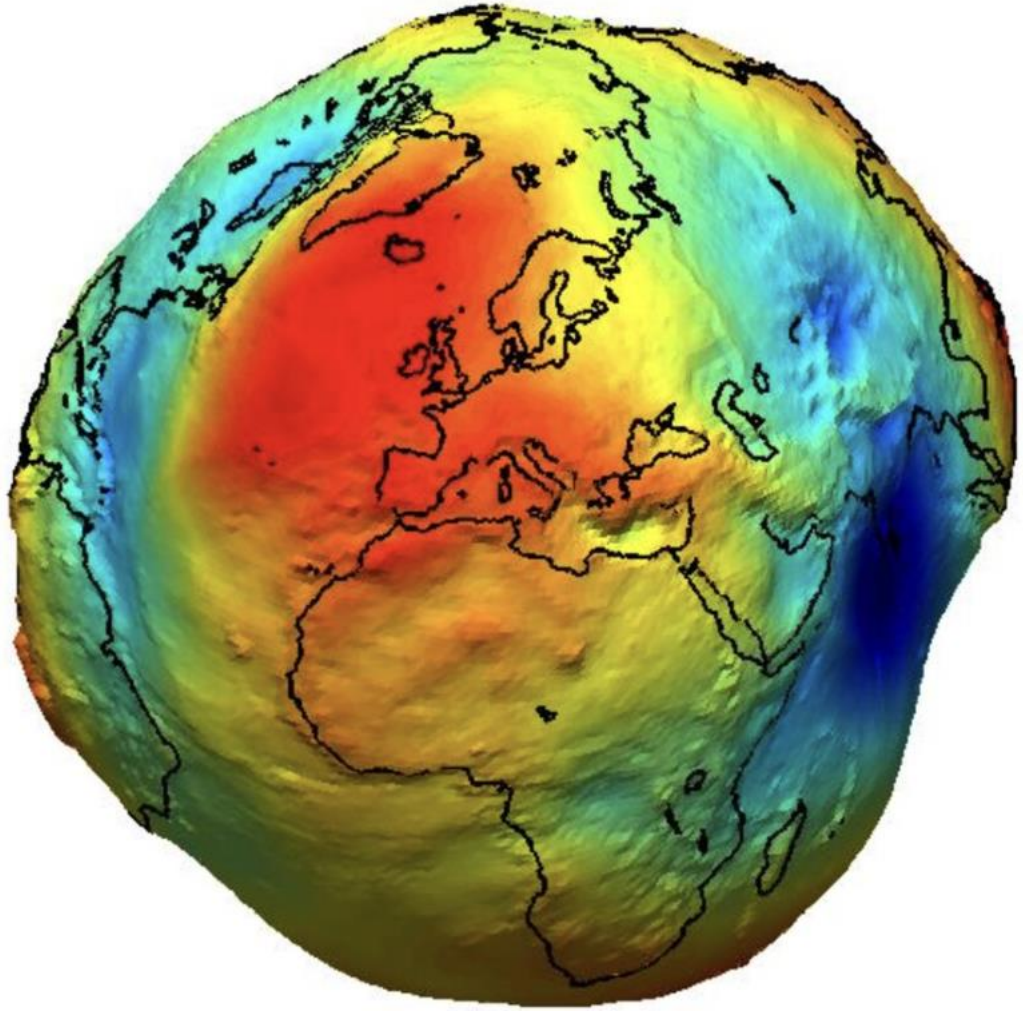
Spatial data carry a frame of reference

- Spatial data include numerical information that allows you to position them on the Earth.
- These numbers are part of a coordinate system that provides a frame of reference for your data: to locate features on the surface of the Earth, to align your data with other data, to perform spatially accurate analysis, and to create maps.
- How does this work?

A CRS tells a GIS where the numbers are

- To locate, combine and visualise spatial data accurately, each dataset needs a coordinate reference system (CRS).
- A Geographic Coordinate System (GCS) gives locations as angles: latitude and longitude, in degrees.
- A Projected Coordinate System (PCS) is a GCS plus a map projection: locations are in linear units, usually metres.

The Earth’s true shape is lumpy: the geoid

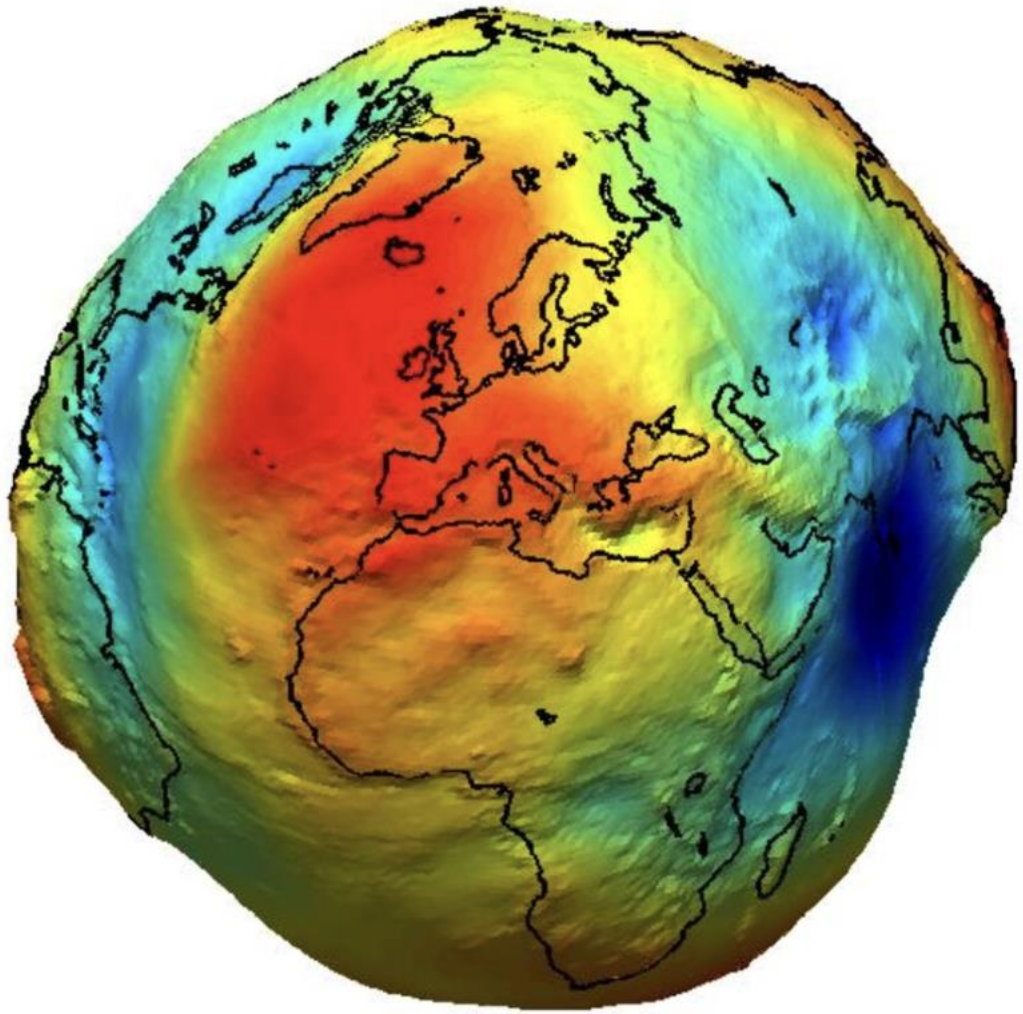




A reference ellipsoid approximates the Earth

- The Earth's true shape, the geoid, is irregular. A mathematical model of it is crucial for a GIS.
- The Earth is not a perfect sphere (which would make calculations much easier) but more closely resembles an ellipsoid.
- In GIS we use a reference ellipsoid as an approximation: a mathematically defined surface that roughly matches the Earth when its topographical features are removed.

Same Earth, simpler shape

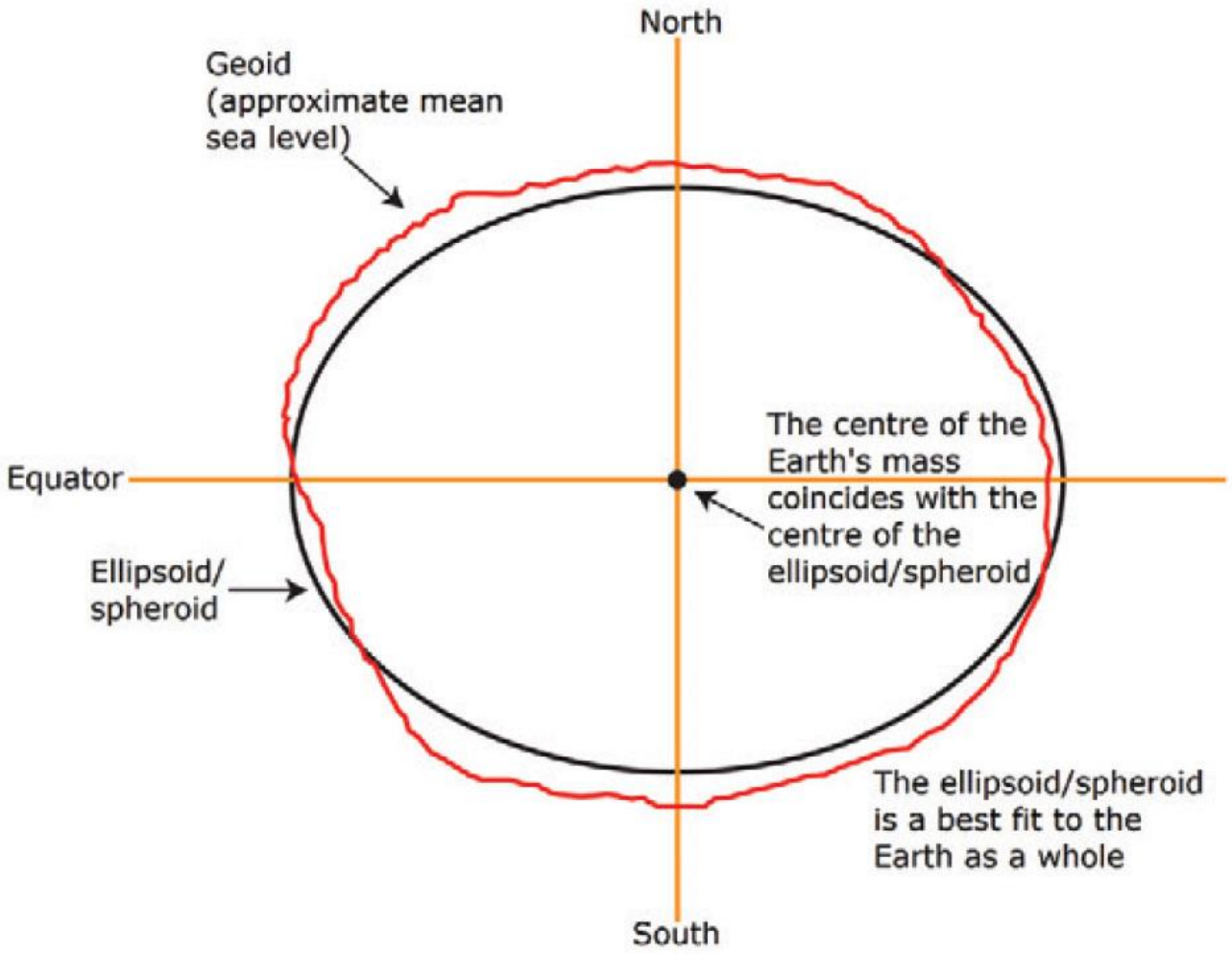




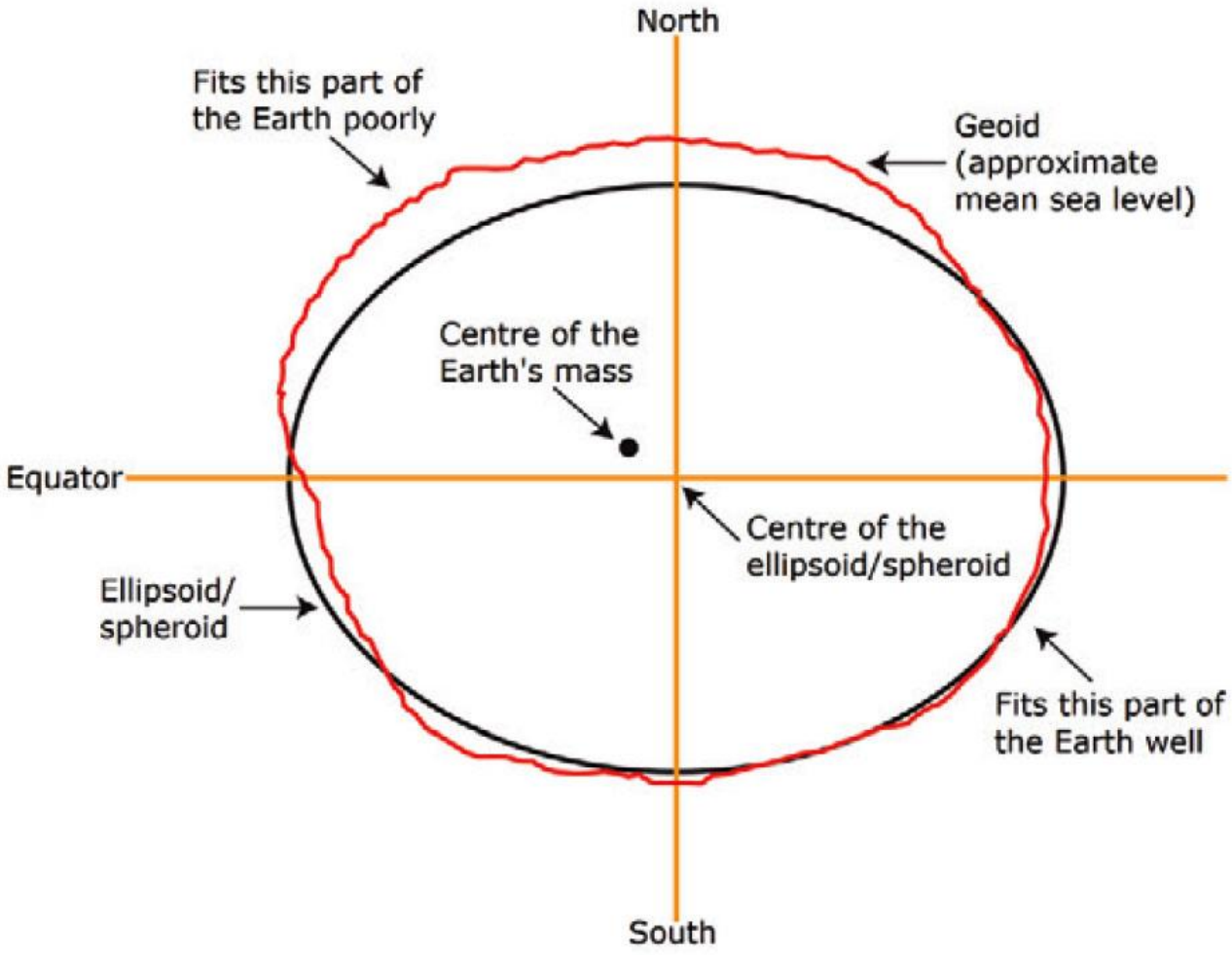
An ellipsoid fits well in some places only

- Once we have a reference ellipsoid, we can align it with the geoid and map the Earth's surface features onto it.
- A reference ellipsoid is an approximation: in some places it matches the geoid closely, in others it deviates significantly.

Global alignment: an acceptable fit everywhere



Local alignment: a close fit in one region



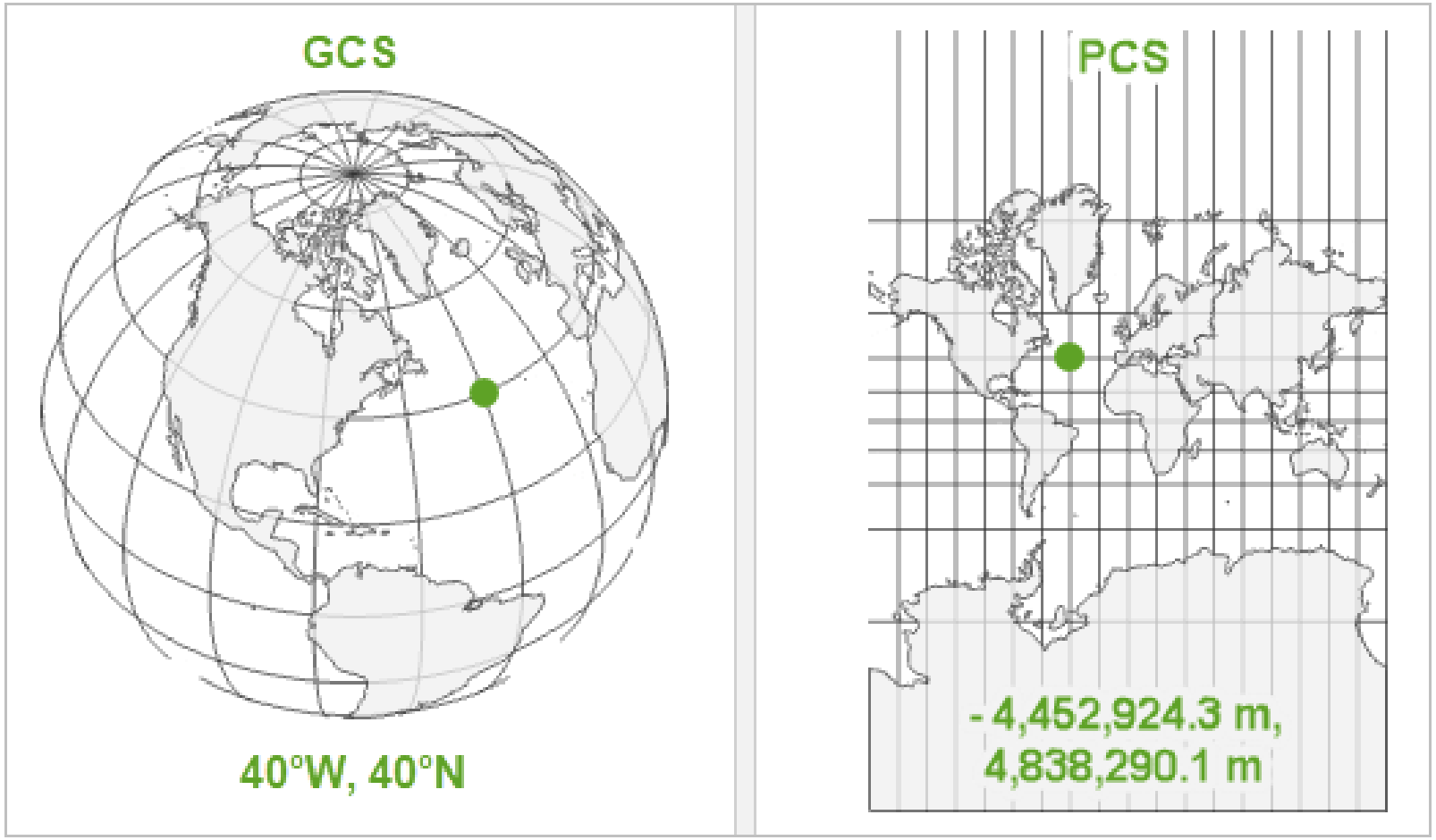
The alignment defines the datum

- The alignment can be local, where the ellipsoid is closely fitted to the geoid at one location, or geocentric, where it is aligned with the centre of the Earth.
- How the ellipsoid is aligned to the geoid defines a datum.
- A datum, together with a prime meridian and angular units, defines a Geographic Coordinate System (GCS).
- A GCS defines where the data are located on the Earth's surface using latitude and longitude.

A GCS uses degrees, a PCS uses metres

- A Geographic Coordinate System (GCS) defines the location of data on the Earth's surface using angular units, usually degrees, because the Earth is round.
- A Projected Coordinate System (PCS) defines how to represent the data on a flat surface, using linear units, usually metres, on a Cartesian plane.

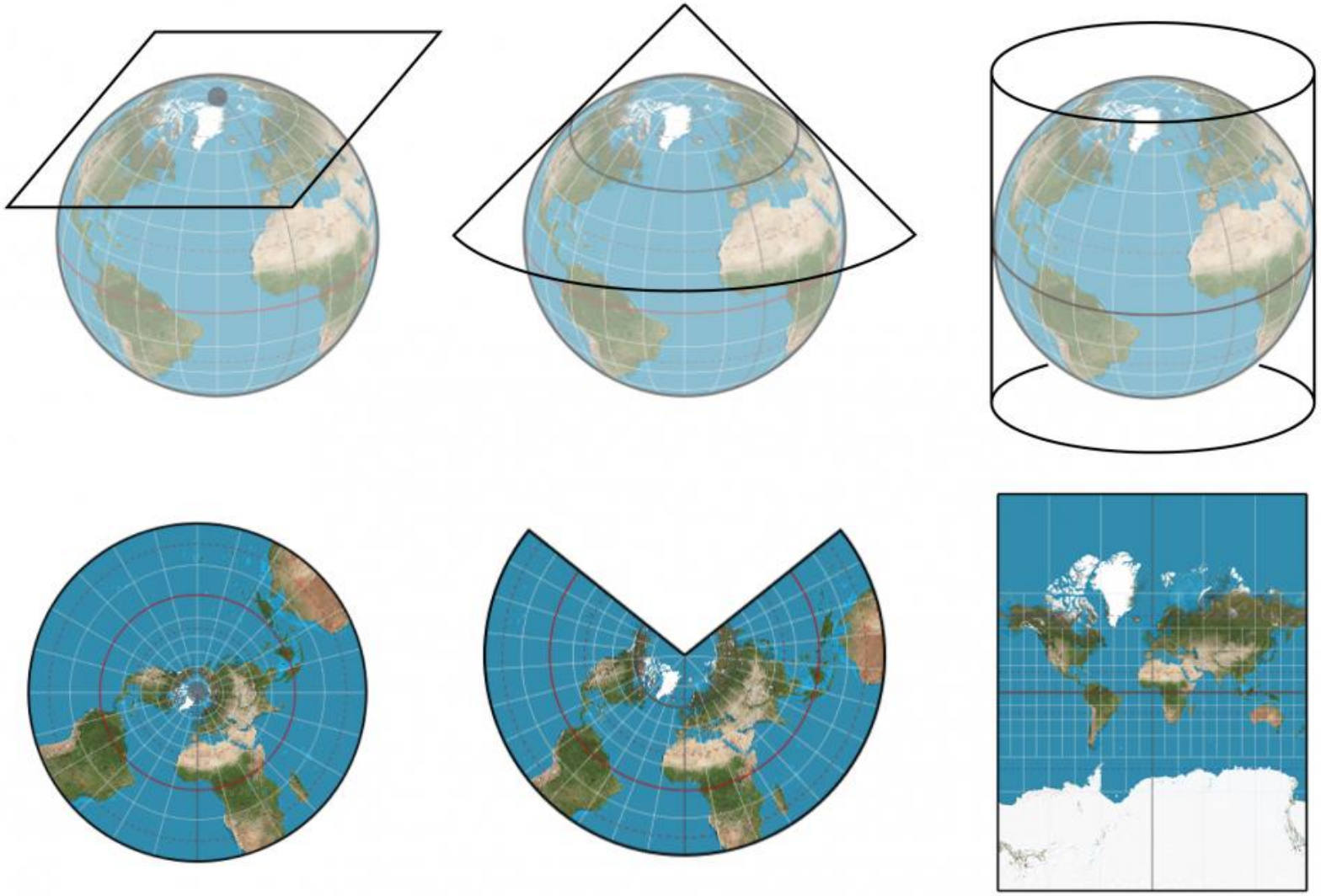
From degrees on a globe to metres on a flat map



A projection turns the globe into a flat map

- A PCS is a reference system for identifying locations and measuring features on a flat (map) surface.
- Going from a GCS to a PCS requires a mathematical transformation: a map projection.
- There are many projections, in three groups: planar, conical and cylindrical.

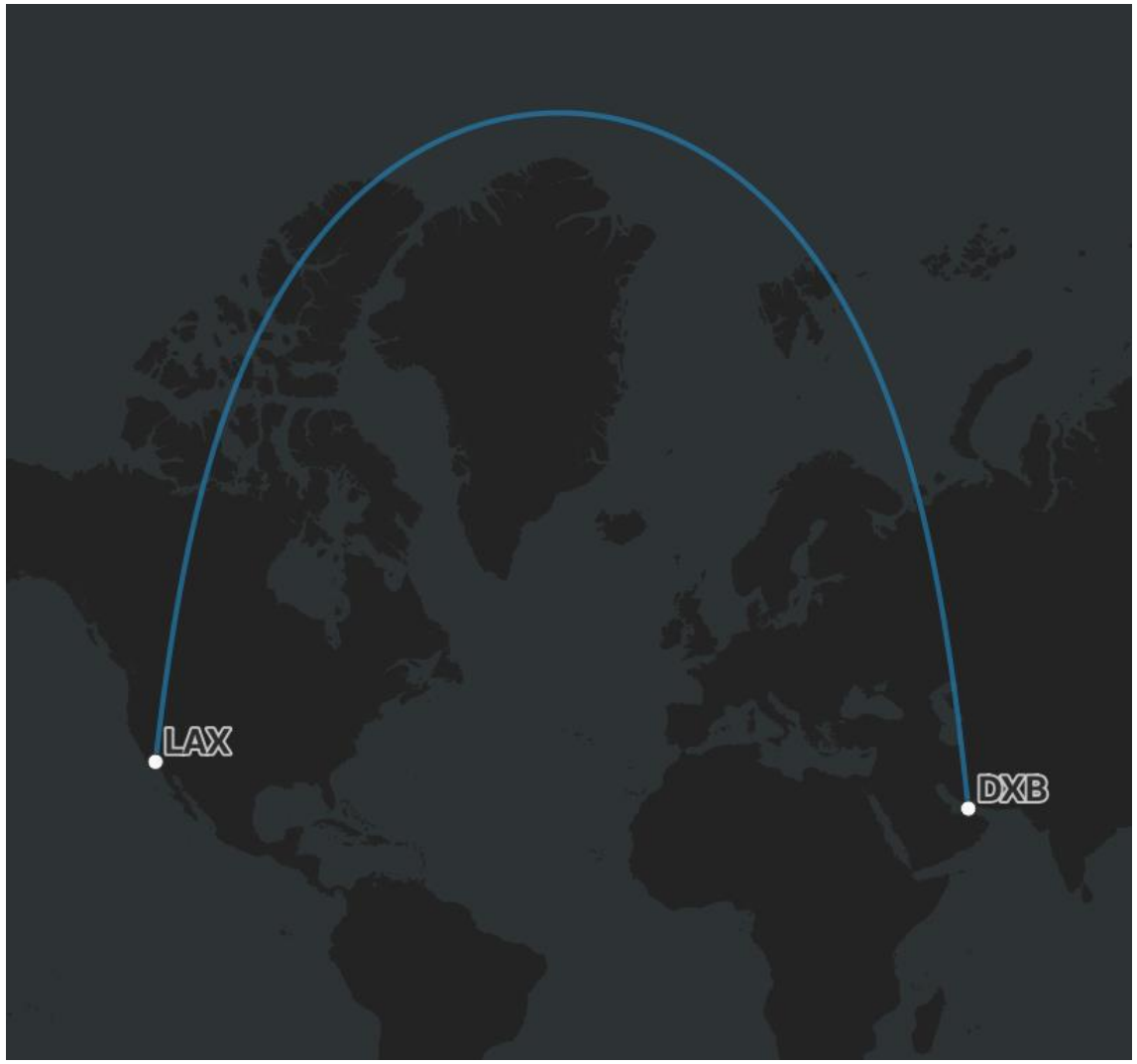
Planar, conical and cylindrical projections



Every projection distorts something

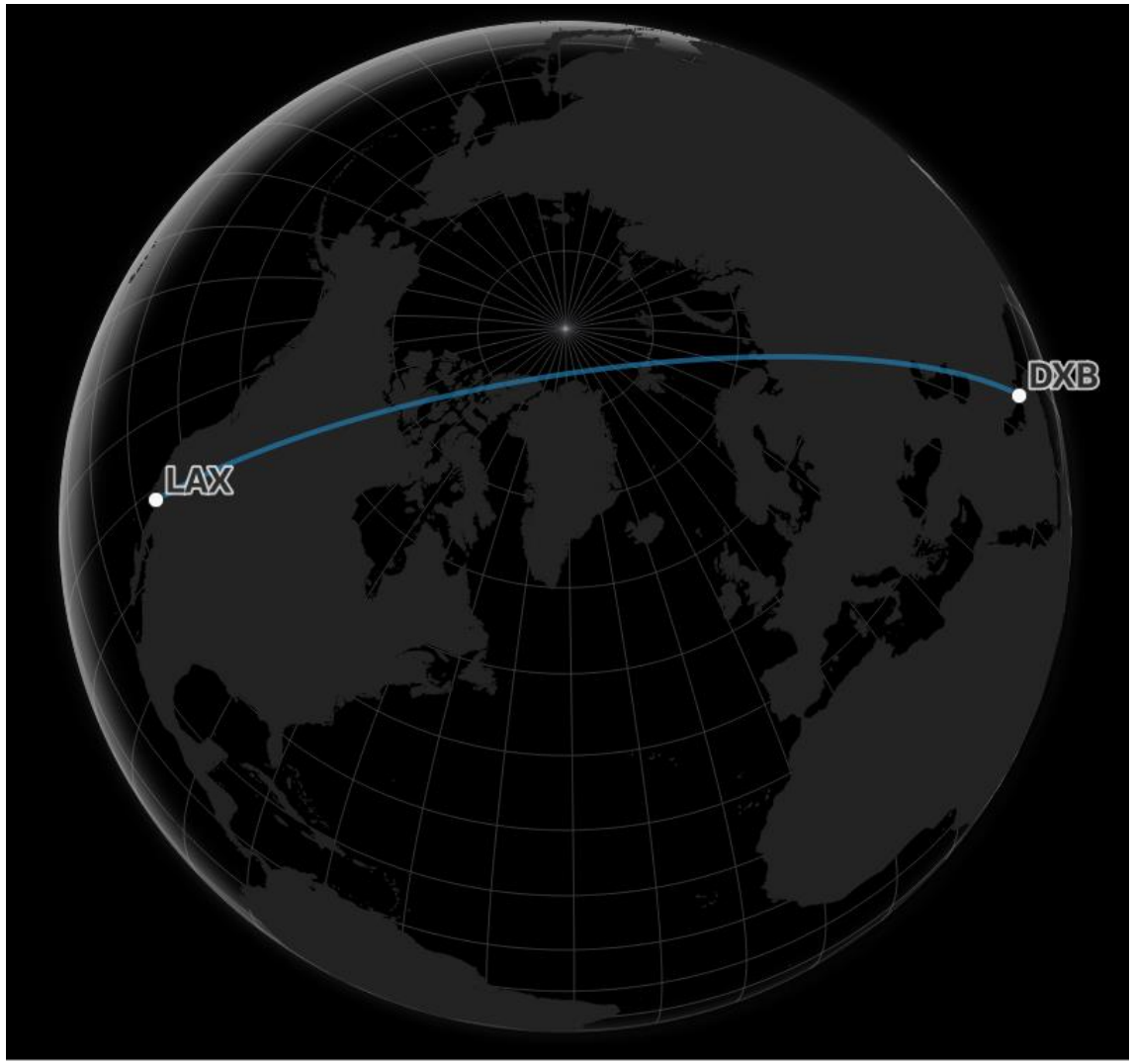
- All projections distort real-world geographic features in terms of shape, area, distance or direction.
- Each map projection preserves only one or two of these four properties well.

On Mercator, the shortest flight looks curved



Alasdair Rae. *https://automaticknowledge.co.uk/*

On a globe, the same route is straight

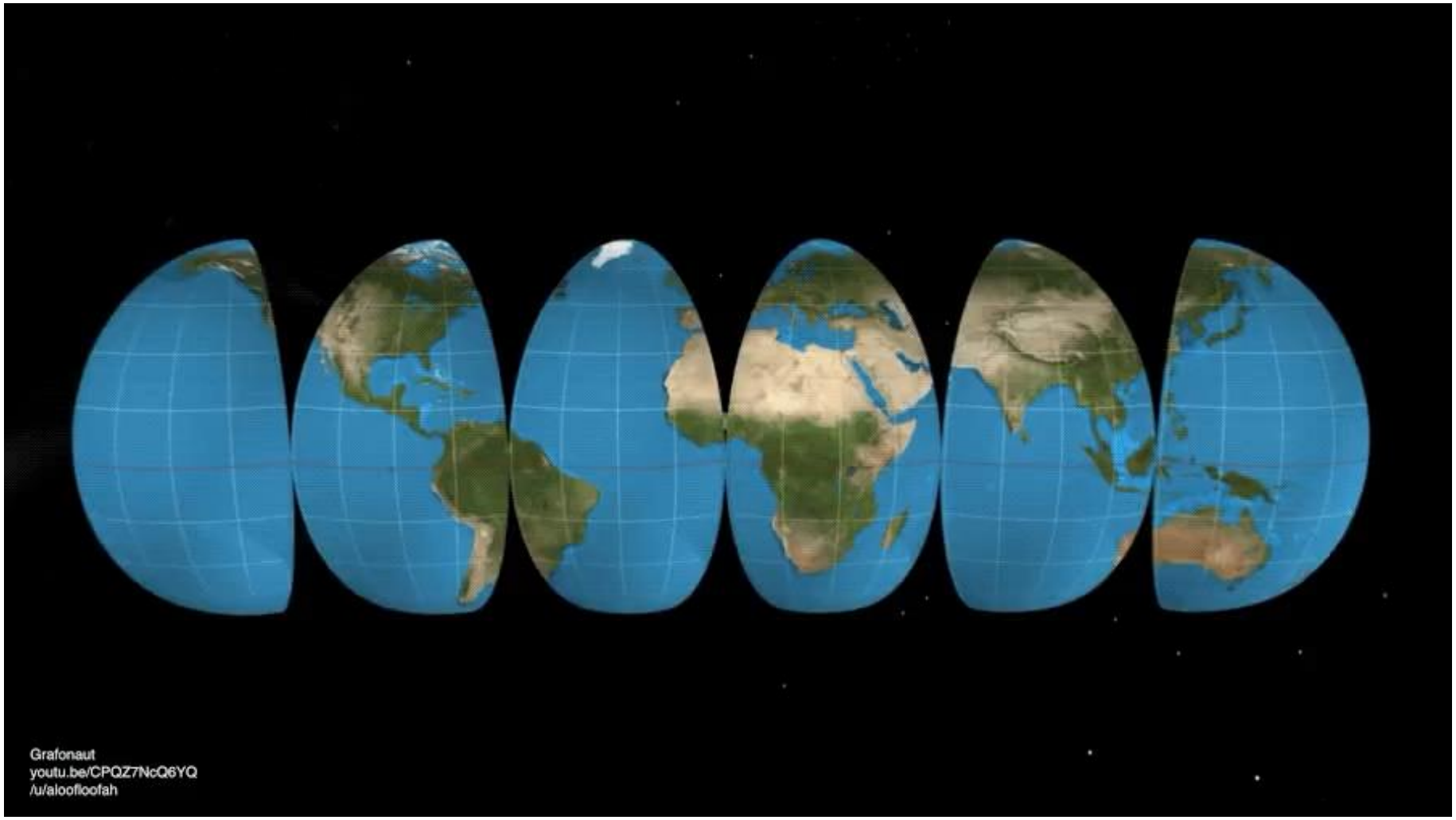


Alasdair Rae. *https://automaticknowledge.co.uk/*

Mercator preserves angles, not area

- Probably the most (in)famous projection for the world on a flat surface is the Mercator projection. It is often used on the web.
- It preserves angles, but exaggerates the size of areas near the poles and understates those near the equator.

Other projections distort differently



Explore the distortion yourself

- Effective visualisation of the effects of the Mercator projection on individual countries: https://thetruesize.com/

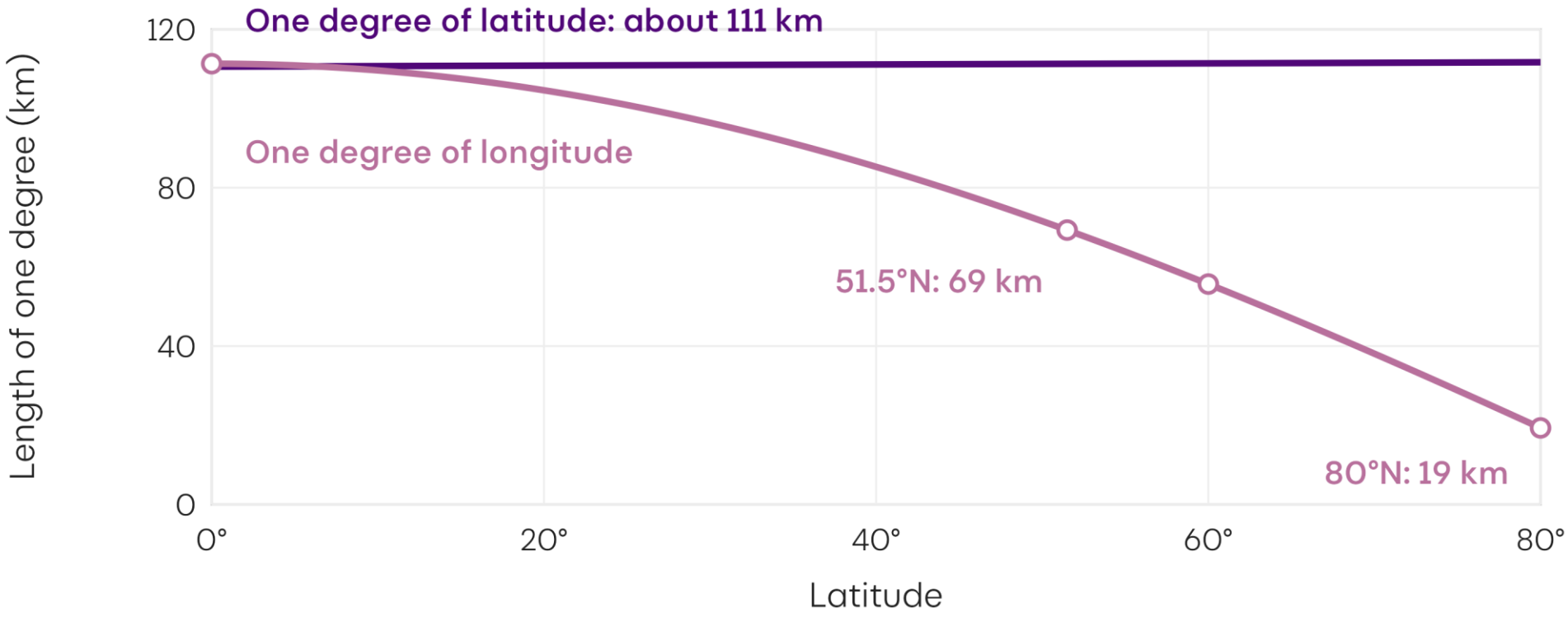
Choosing and checking a CRS

- WGS 84 is a global GCS (latitude and longitude, used by GPS). It suits global data, but not measuring distances or areas.
- The British National Grid is a PCS for Great Britain: a Transverse Mercator projection on the OSGB36 datum, with locations in metres. It is like UTM, but one national grid without zone boundaries.
- Large countries or regions may need different projections.
- Existing data usually have a CRS assigned. Layers that are combined must share one.

EPSG codes name a coordinate reference system

- The details of a CRS are often given as an EPSG code. The EPSG registry is a public registry of geodetic datums, spatial reference systems, ellipsoids, transformations and units.
- EPSG:4326: WGS 84, latitude and longitude in degrees (GPS)
- EPSG:27700: British National Grid, metres
- EPSG:3857: Web Mercator, used by web maps
- Transforming data from one CRS to another is called reprojecting.

Degrees are angles, not lengths



Checking and changing a CRS

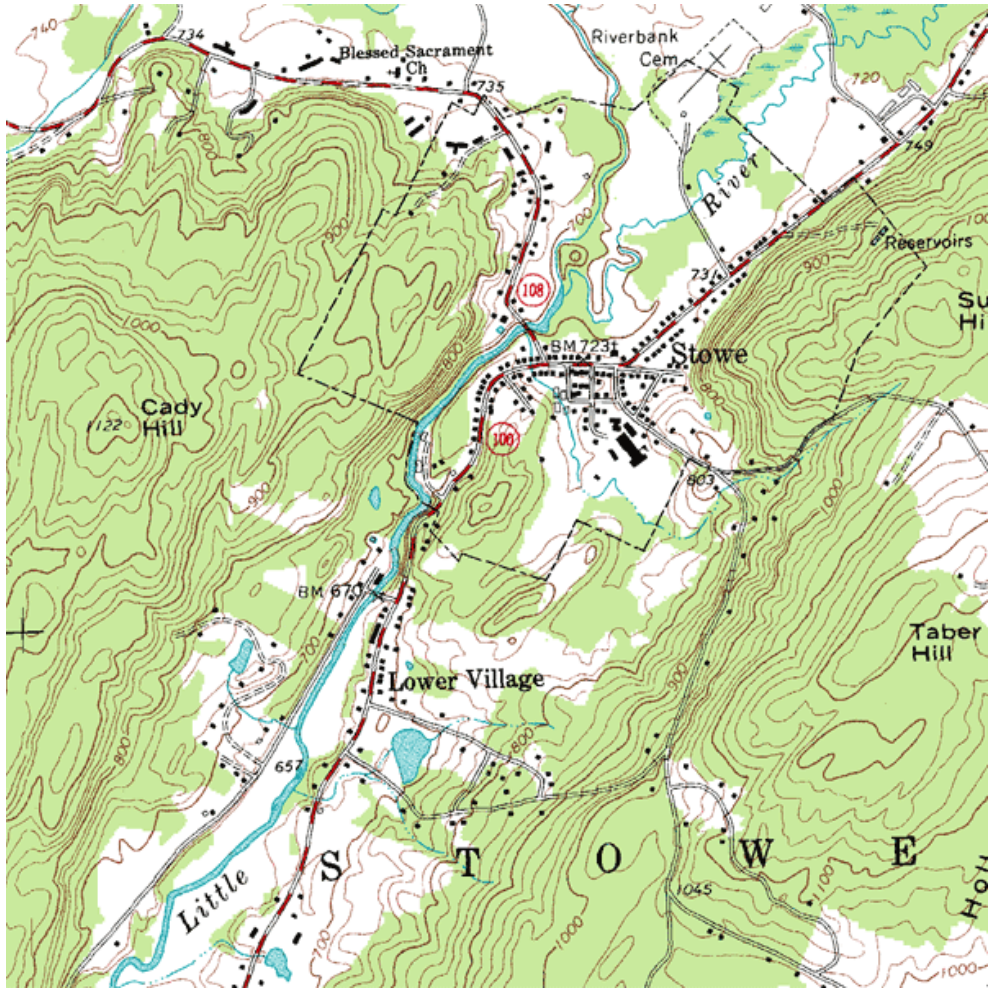
- Look at the CRS of every layer before you combine them, and check its units.
- If the CRS is missing, find it in the metadata. Do not guess.
- Setting a CRS only changes the label. Reprojecting changes the coordinates.
- Measure distances and areas in a projected CRS in metres, for example EPSG:27700 in Great Britain.
- Reprojecting a raster resamples its cells: choose the method that suits the data type (week 6).

Making maps

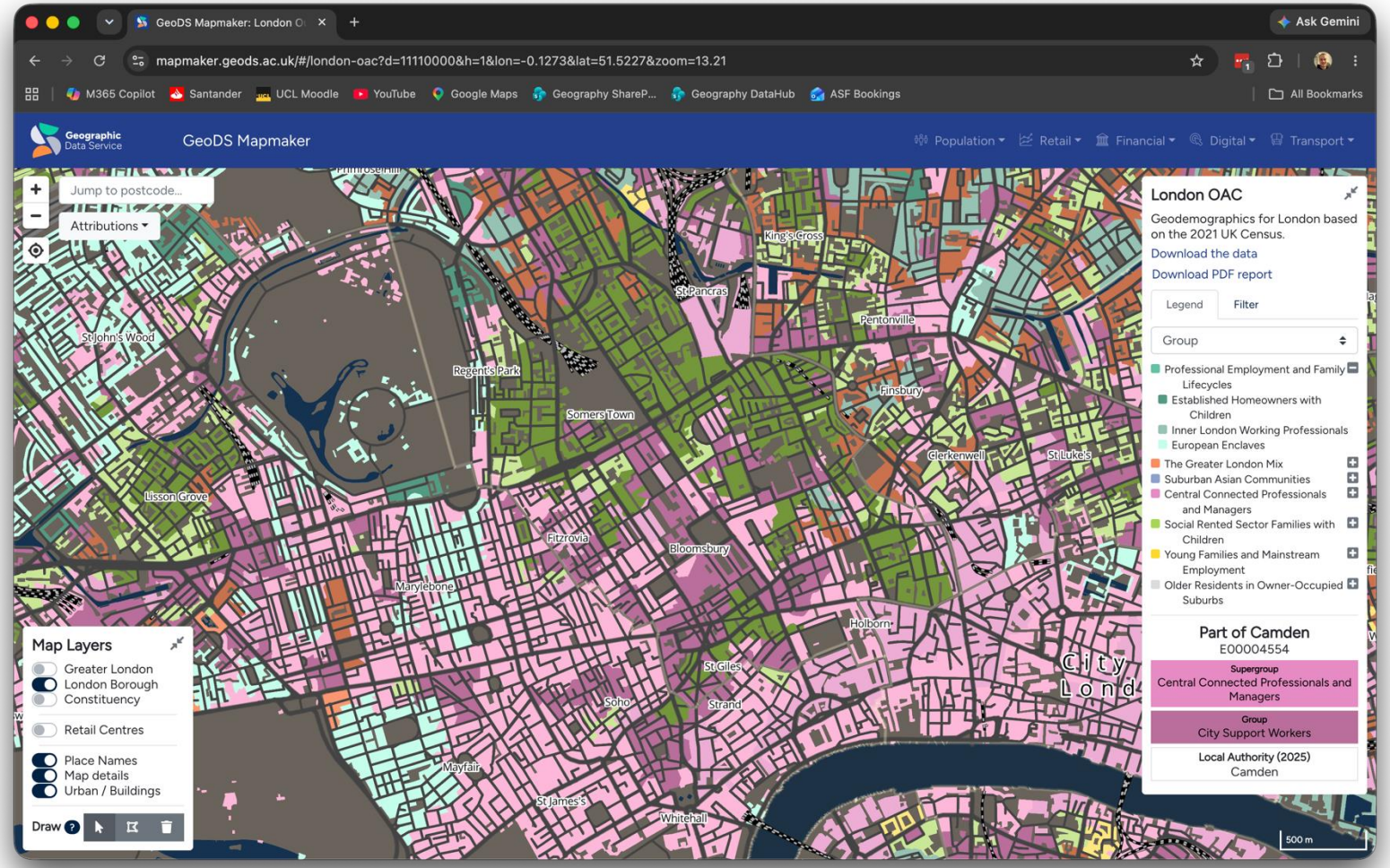
Topographic and thematic maps

- Topographic maps: indicate features of the land's surface (e.g. mountains, hills, valleys, rivers).
- Thematic maps: graphical outputs that typically show geographic patterns of a particular theme in a geographic area.

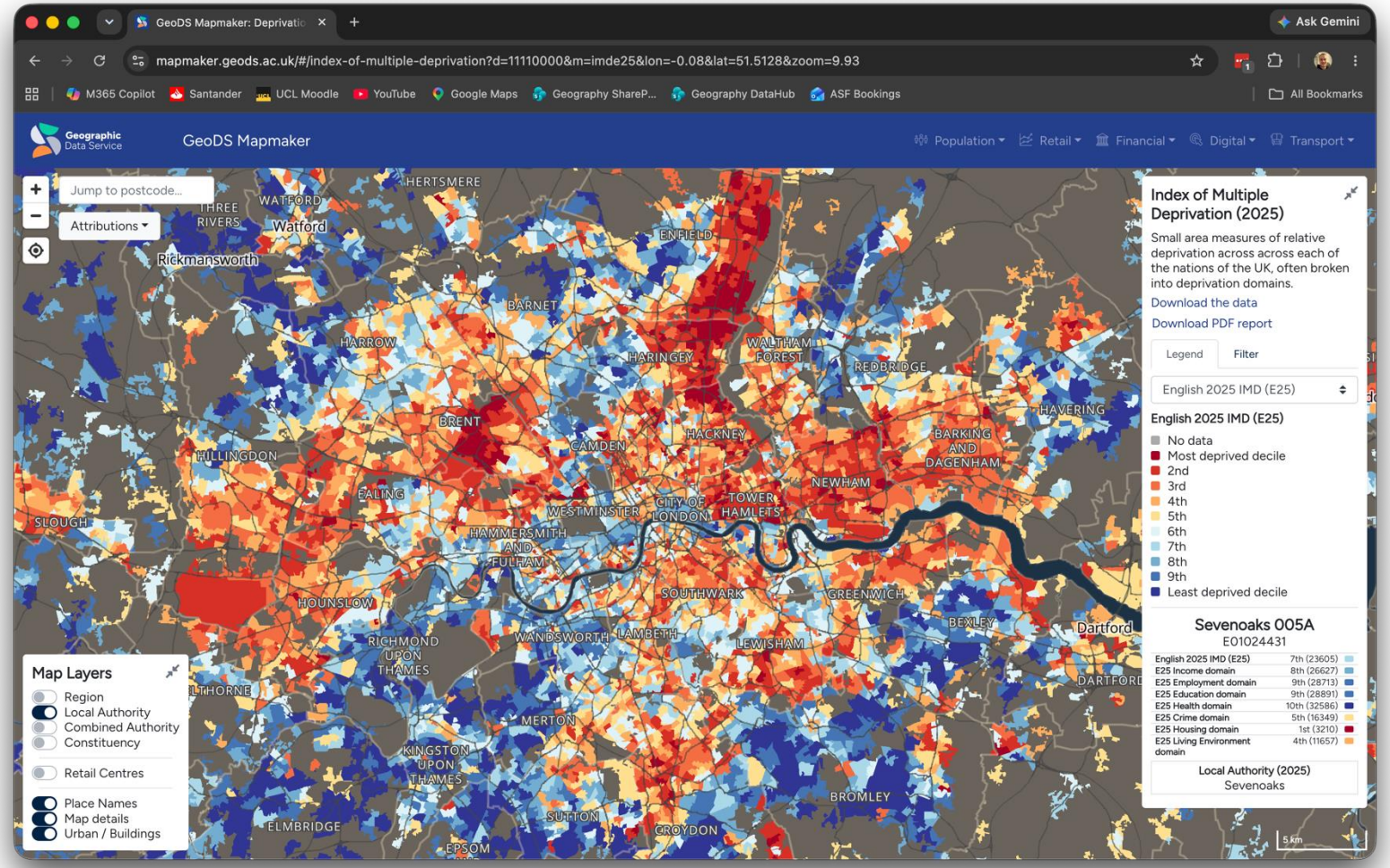
Topographic maps show the land surface



Thematic: London Output Area Classification



Thematic: Index of Multiple Deprivation



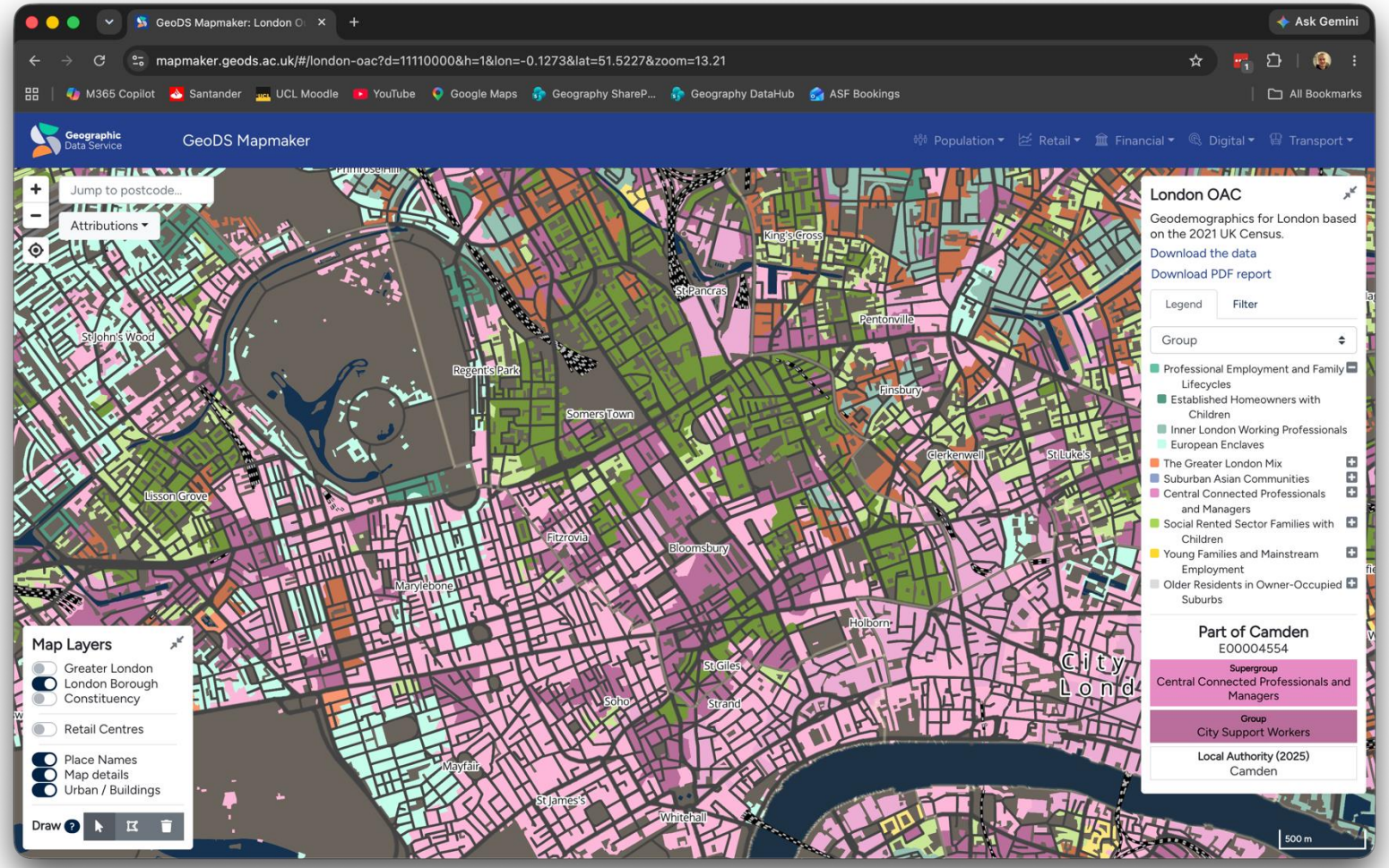
Choropleth maps shade areas by value

- A choropleth map is a thematic map in which areas are shaded or patterned in proportion to the value of the statistical variable shown.
- It can show two types of data: categorical or continuous.

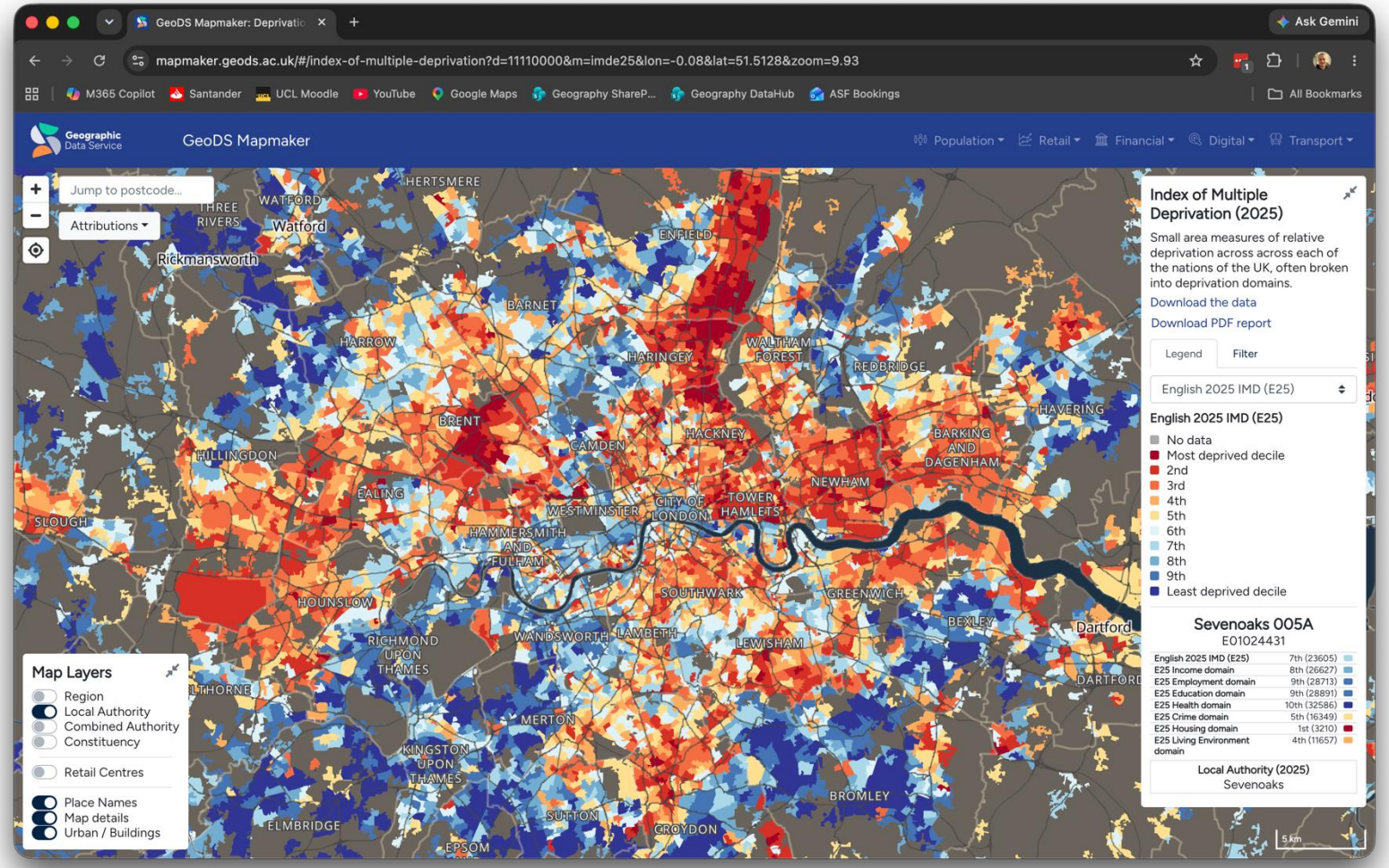
Categorical data: one colour per category

- A separate colour for each unique value, no gradient.
- Descriptive legends help to interpret the colours.
- Colours can be ordered to indicate rank for ordinal data.

Nominal: groups without an order



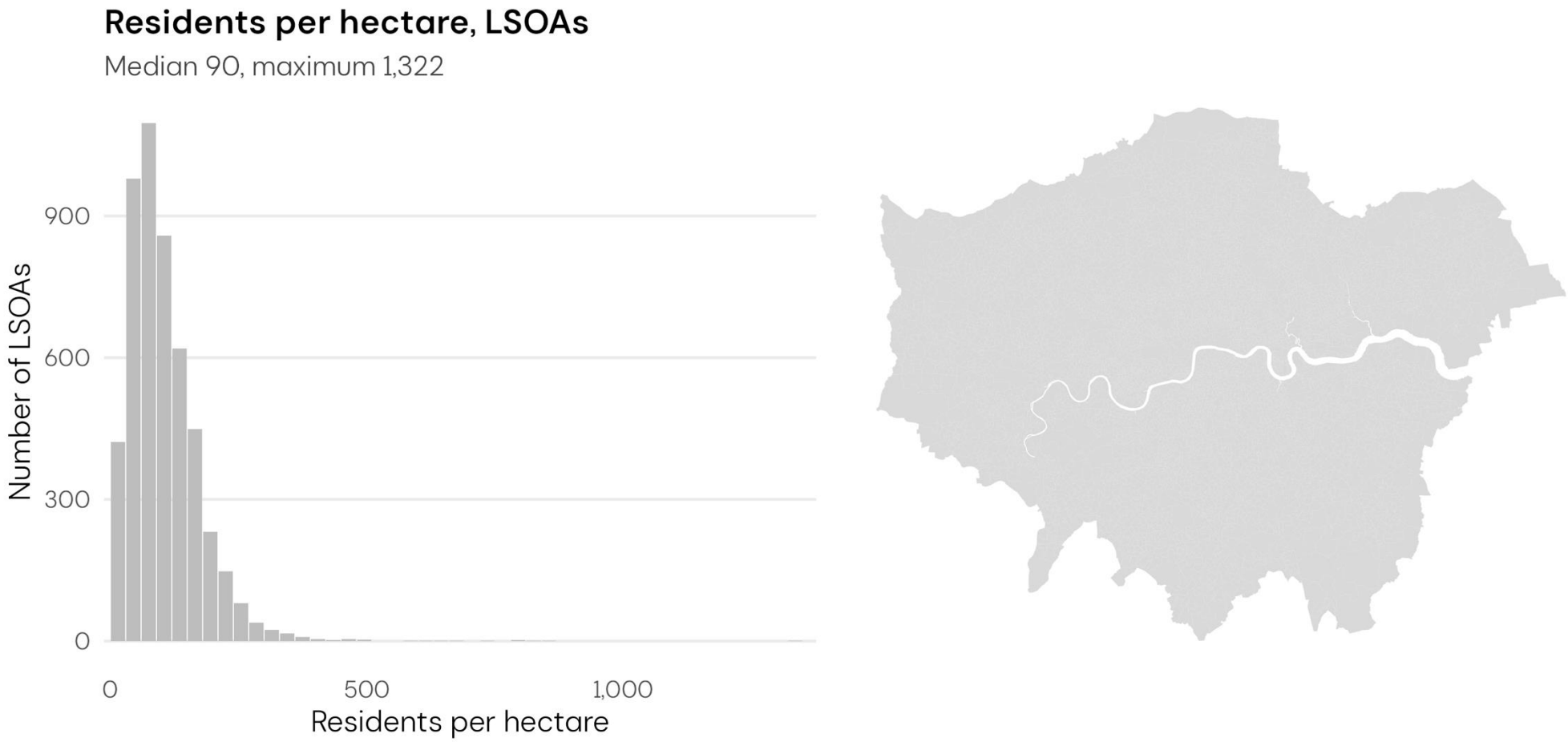
Ordinal: groups with an order



Continuous data are divided into classes

- Continuous variables are divided into classes, the spatial equivalent of the bins of a histogram.
- The number of classes depends on the data; there is no single right number.
- The distribution of the data, shown in a histogram, helps to choose where to set the breaks.
- There is always a trade-off between information loss and readability.

Start with the distribution

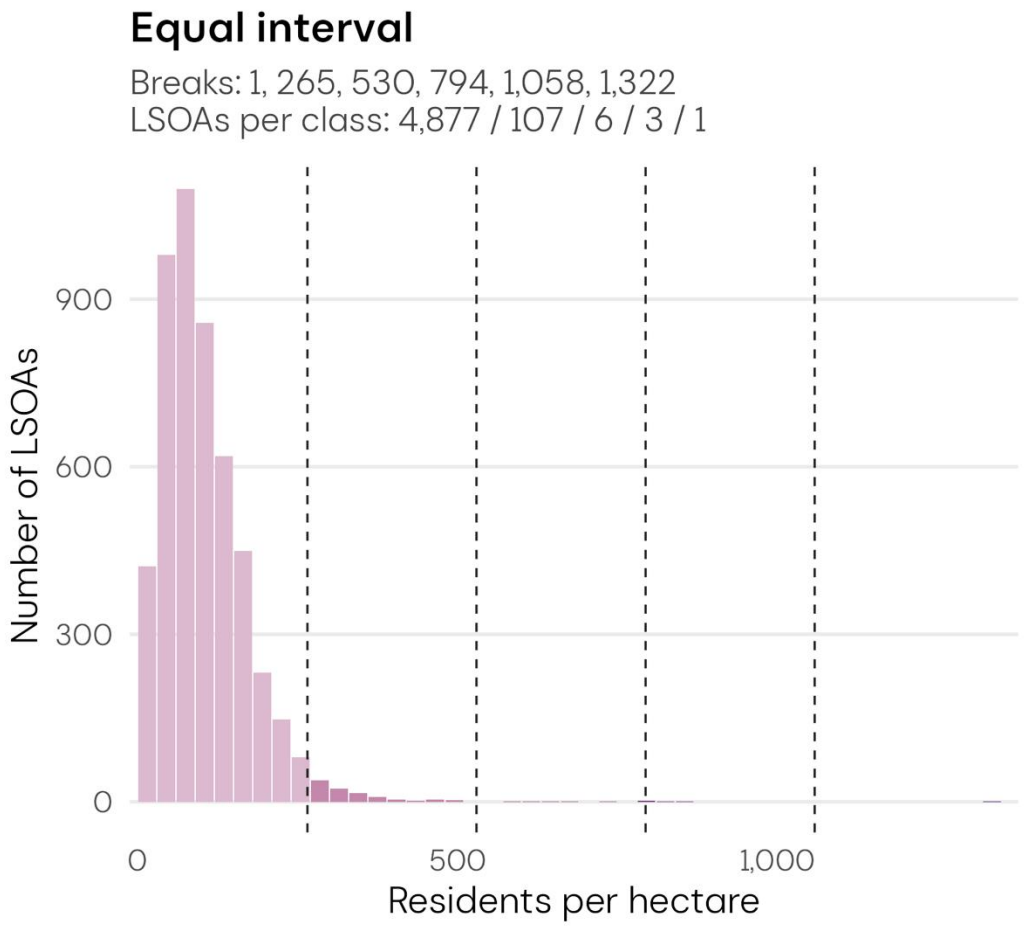


Source: ONS, Census 2021. Contains OS data © Crown copyright and database right 2021.

Equal interval: classes of equal width

- Take the range of the data and split it into classes of equal width.
- Splits are based on the numerical value.
- With skewed data, most areas fall in the first class and a few outliers are spread over the rest.

Equal interval on skewed data



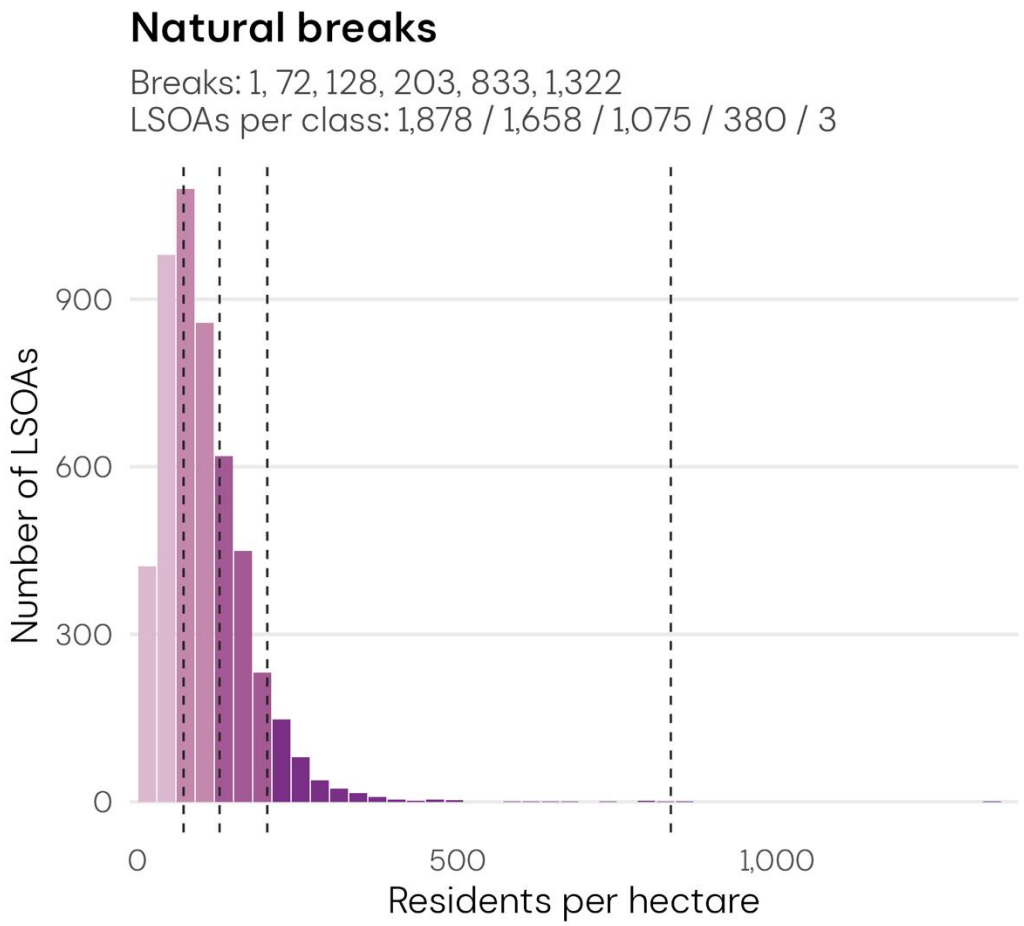


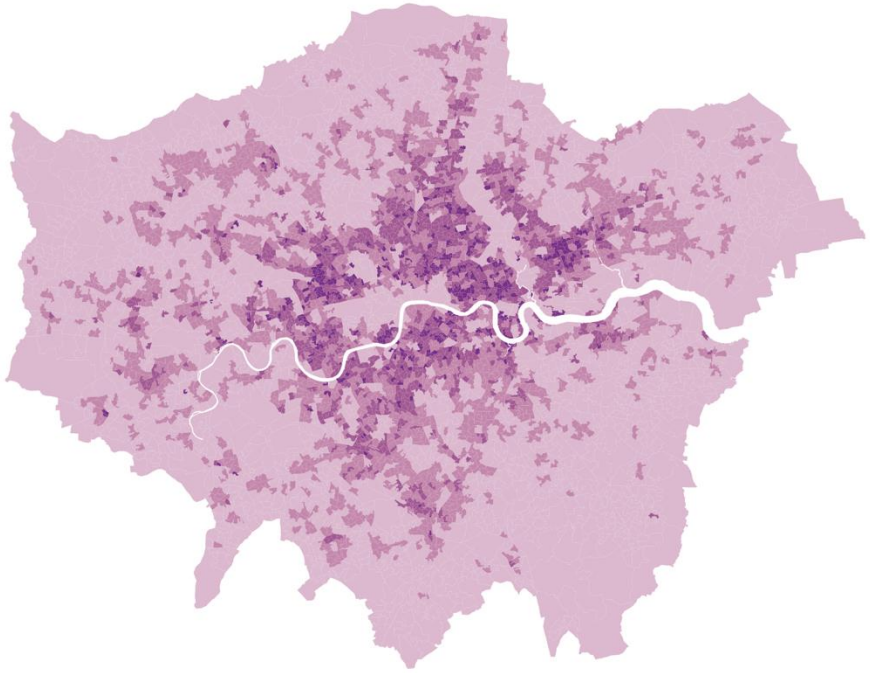
Source: ONS, Census 2021. Contains OS data © Crown copyright and database right 2021.

Natural breaks: classes follow the clusters

- Splits are chosen to minimise the variation within each class and maximise the variation between classes.
- Splits are based on clustering in the data.
- With skewed data, outliers end up in their own class, depending on the number of classes.

Natural breaks on skewed data



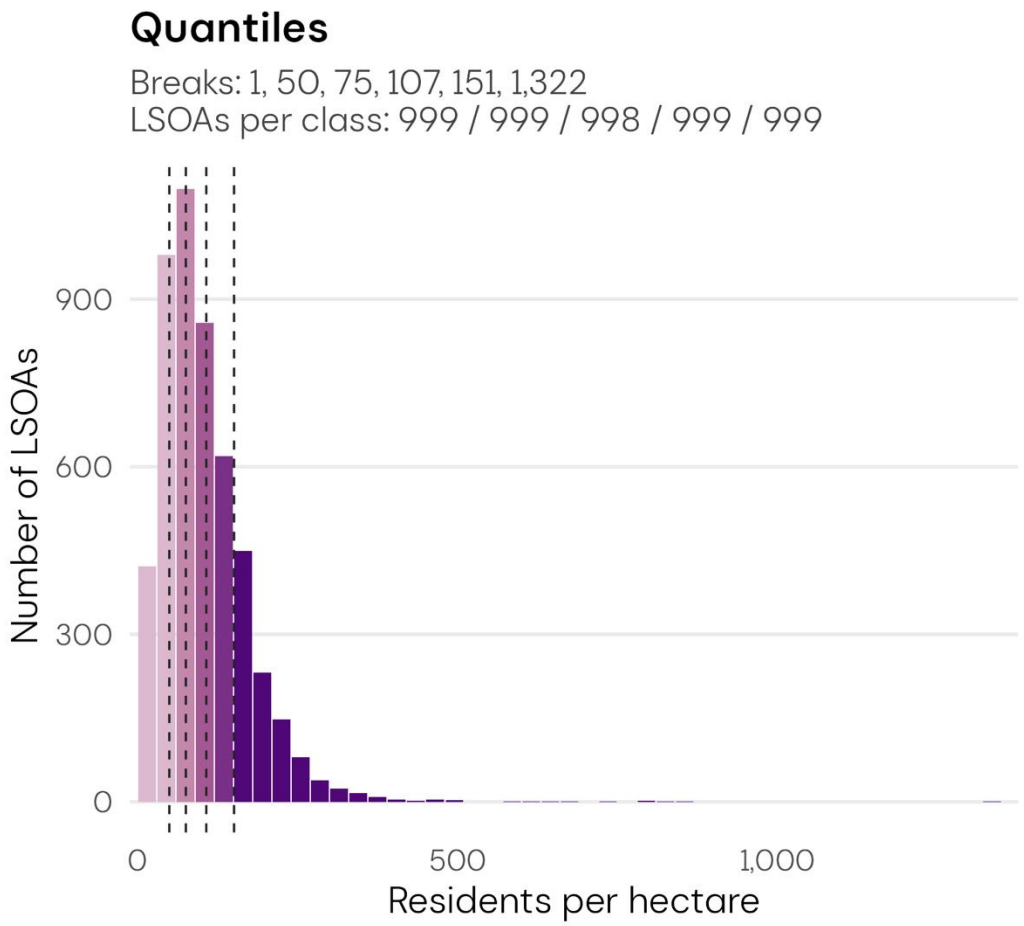


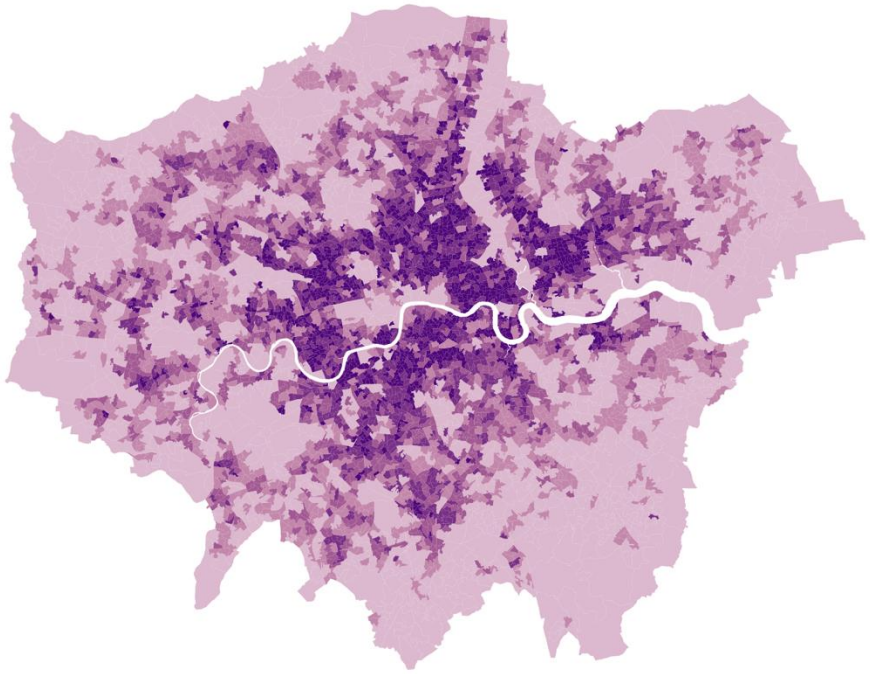
Source: ONS, Census 2021. Contains OS data © Crown copyright and database right 2021.

Quantiles: the same number of areas per class

- Split the distribution so that each class holds the same number of areas.
- Splits are based on the rank of the value.
- With skewed data, very different values can end up in the same class.

Quantiles on skewed data



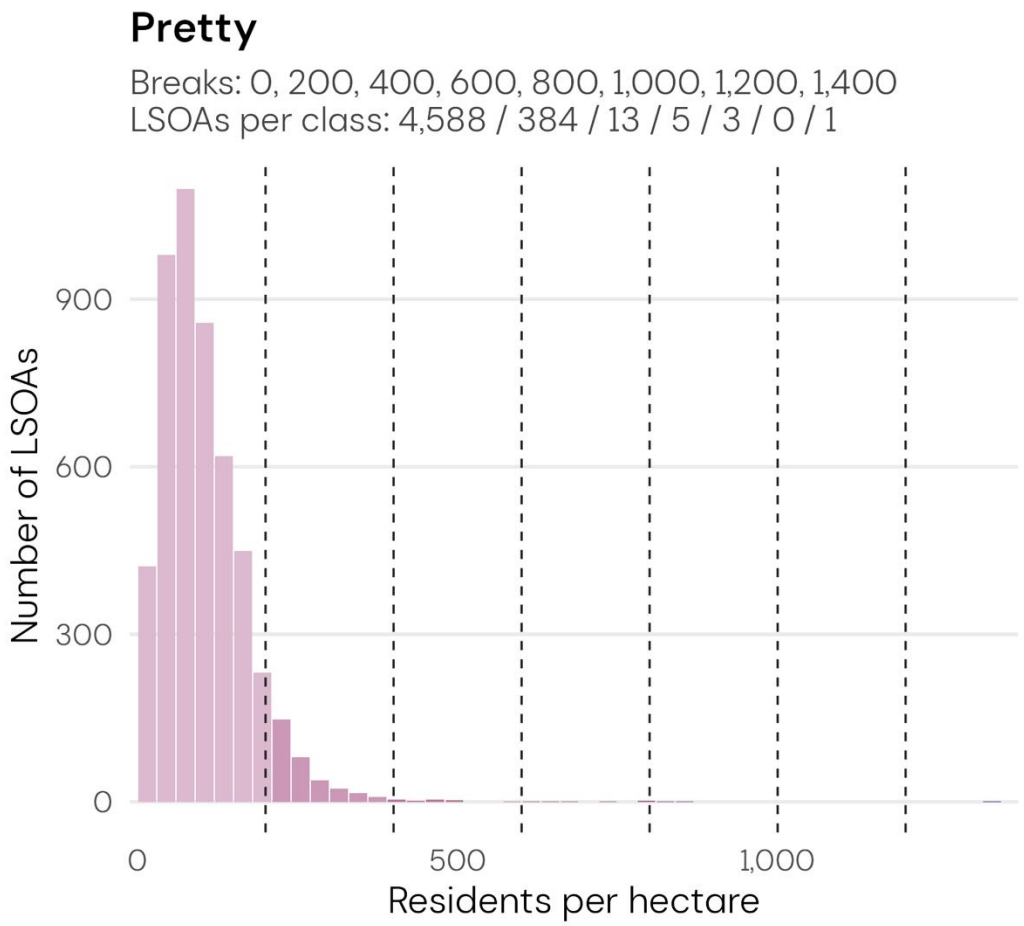


Source: ONS, Census 2021. Contains OS data © Crown copyright and database right 2021.

Pretty: rounded breaks

- Split the distribution using a series of rounded values.
- Splits are not based on the data or their distribution.
- With skewed data, some classes hold very few areas, or none at all.

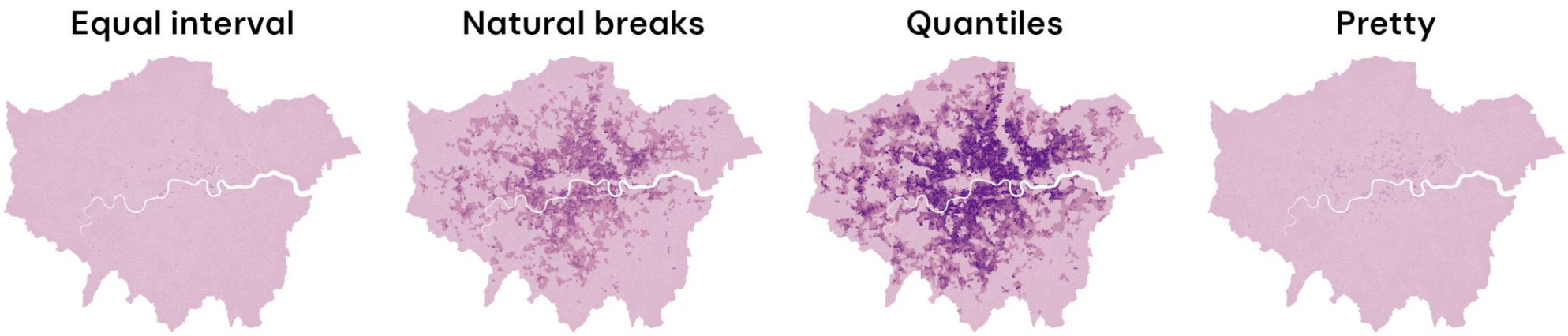
Pretty breaks on skewed data





Source: ONS, Census 2021. Contains OS data © Crown copyright and database right 2021.

Same data, four classifications, four maps

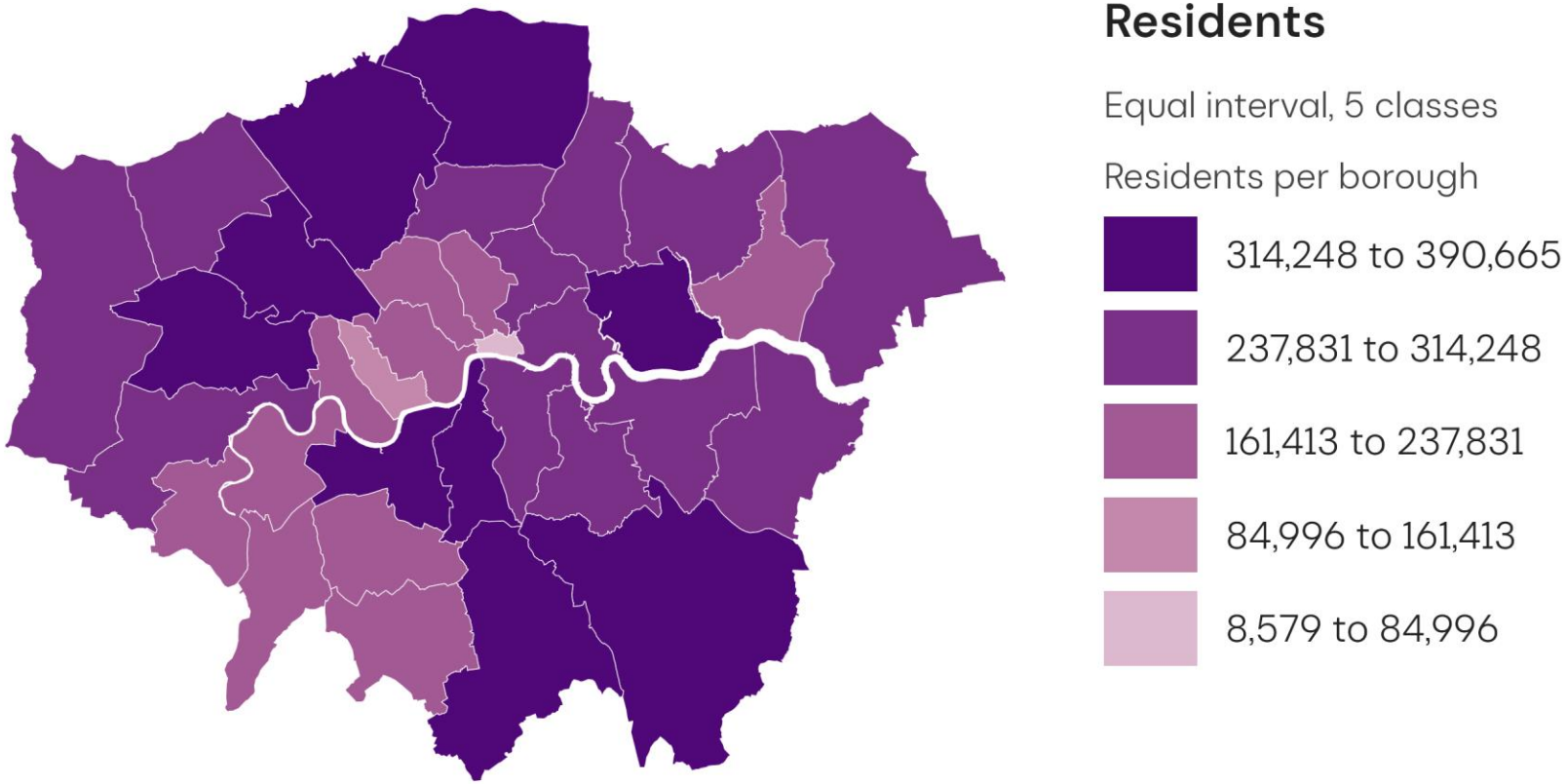


Source: ONS, Census 2021. Contains OS data © Crown copyright and database right 2021.

Standardise counts before mapping

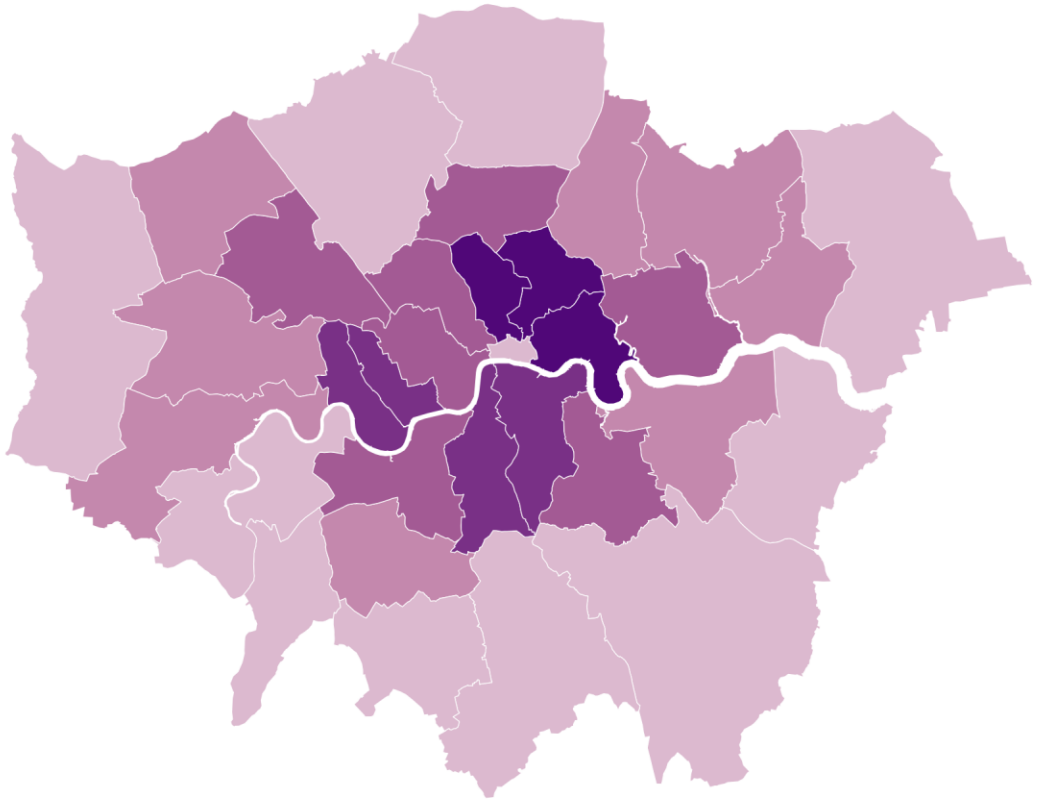
- Plotting raw counts in choropleth maps should generally be avoided, as it can mislead.
- Standardise by dividing the variable of interest (the numerator) by a standardising variable (the denominator), such as total population, area or another relevant factor.

Counts mainly show how big the areas are



Source: ONS, Census 2021. Contains OS data © Crown copyright and database right 2021.

Residents per hectare shows where people live



Residents per hectare

Equal interval, 5 classes

Residents per hectare

- 130 to 157
- 103 to 130
- 76 to 103
- 49 to 76
- 22 to 49

Source: ONS, Census 2021. Contains OS data © Crown copyright and database right 2021.

Colour scales should match the data

Sequential

Ordered, low to high



low high

Diverging

A meaningful midpoint



below midpoint above

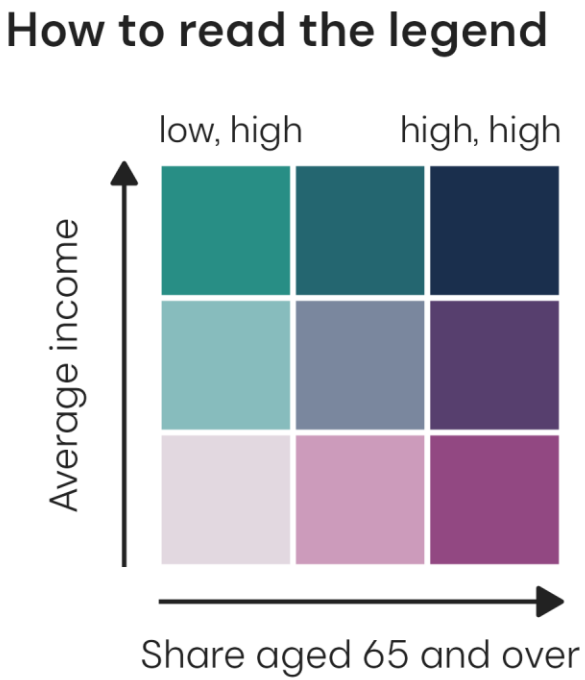
Qualitative

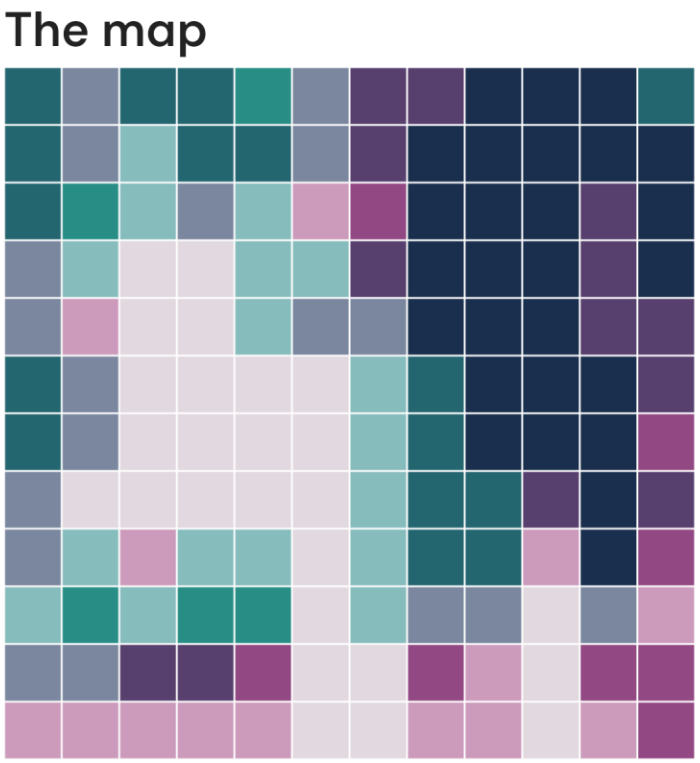
Categories, no order



category A to F

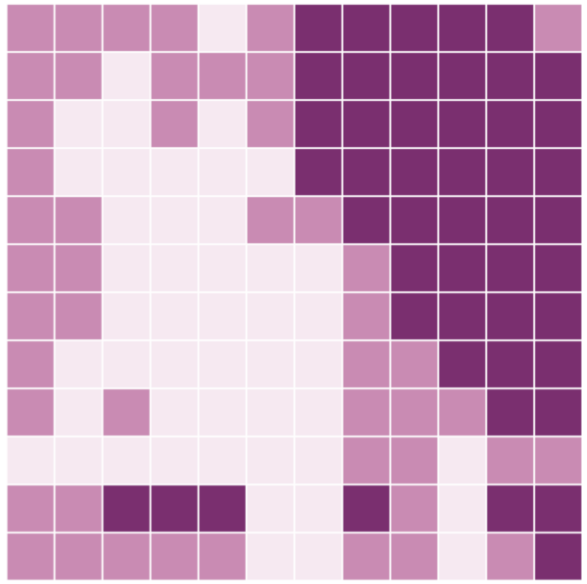
A bivariate map shows two variables at once





Classify each variable, then combine

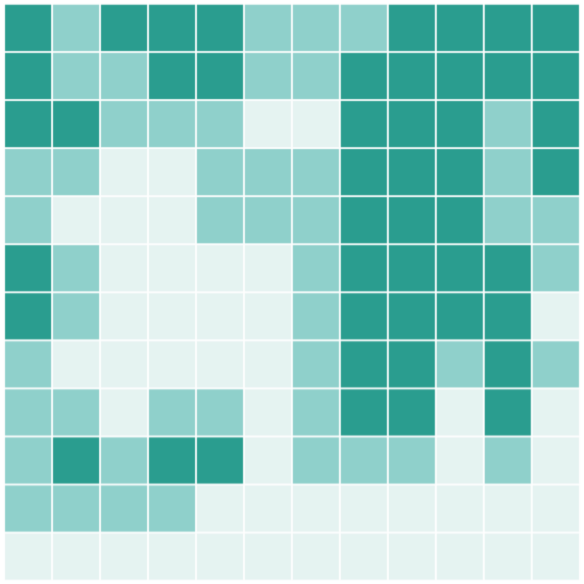
Classify x



3 classes of x

+

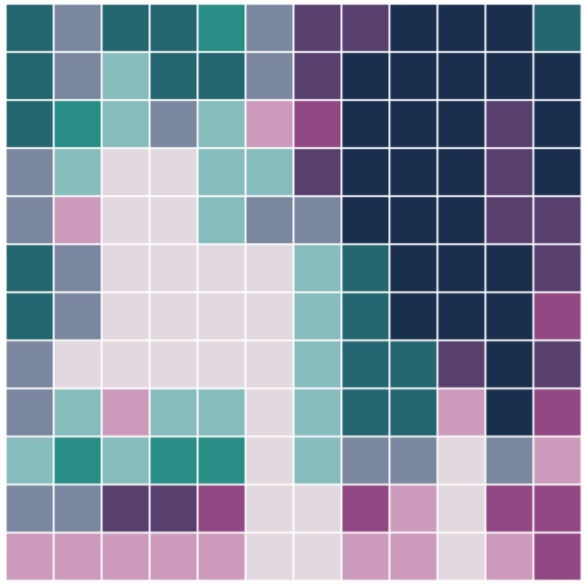
Classify y



3 classes of y

=

Combine the two



3 by 3 = 9 classes

Every map needs these elements

All maps should have a:

- Title
- Legend
- Scale bar / scale text
- North arrow
- And: attribution and data sources

Optional:

- Text and labels – sometimes important for context, sometimes distracting!

Do not forget the map elements



Effective mapping needs analysis and design

- Effective mapping requires a combination of good analysis and good visualisation.

  GIGO: Garbage In, Garbage Out.
- Avoid oversimplification or unnecessary detail.

A tool that runs is not a tool that is right

- AI assistants such as Microsoft Copilot write fluent code and confident explanations, also when they are wrong.
- The costly errors raise no error message: a wrong unit, a wrong CRS, a join that adds or drops rows, a default that does not suit the data.
- Ask yourself what the answer should look like before you run the code: a size, a unit, a total, a pattern on the map.
- Check the result against something you trust; explain each step in your own words.
- The responsibility for the result is yours, not the tool's.

Conclusion

Part I

- GCS and PCS link spatial layers to the Earth, and layers must share a CRS.

Part II

- A good map follows from choices; comparisons need matching boundaries and definitions.

Part III

- A tool that runs can still be wrong: check what it produces.

Questions

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